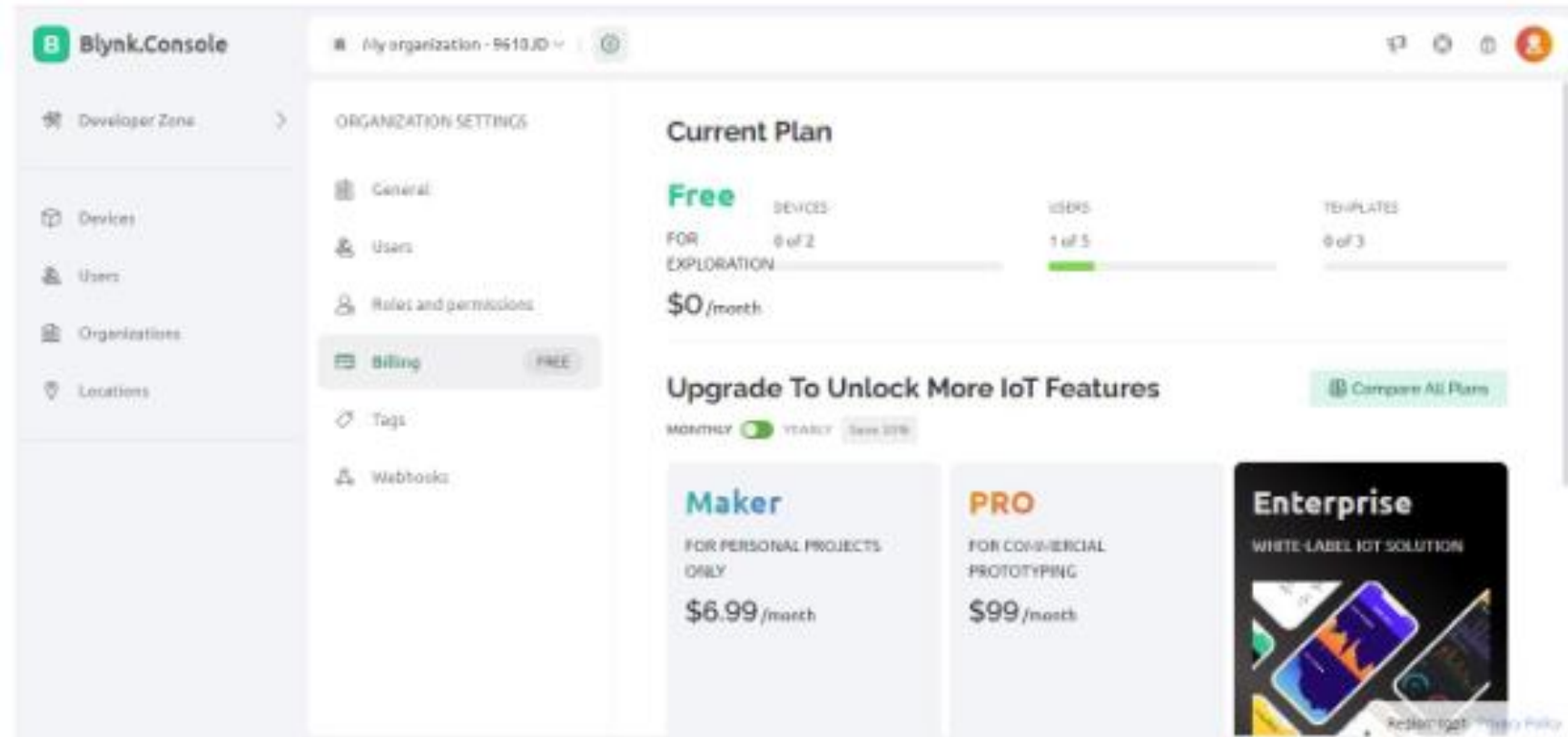


Mobile & IoT

Pertemuan 3

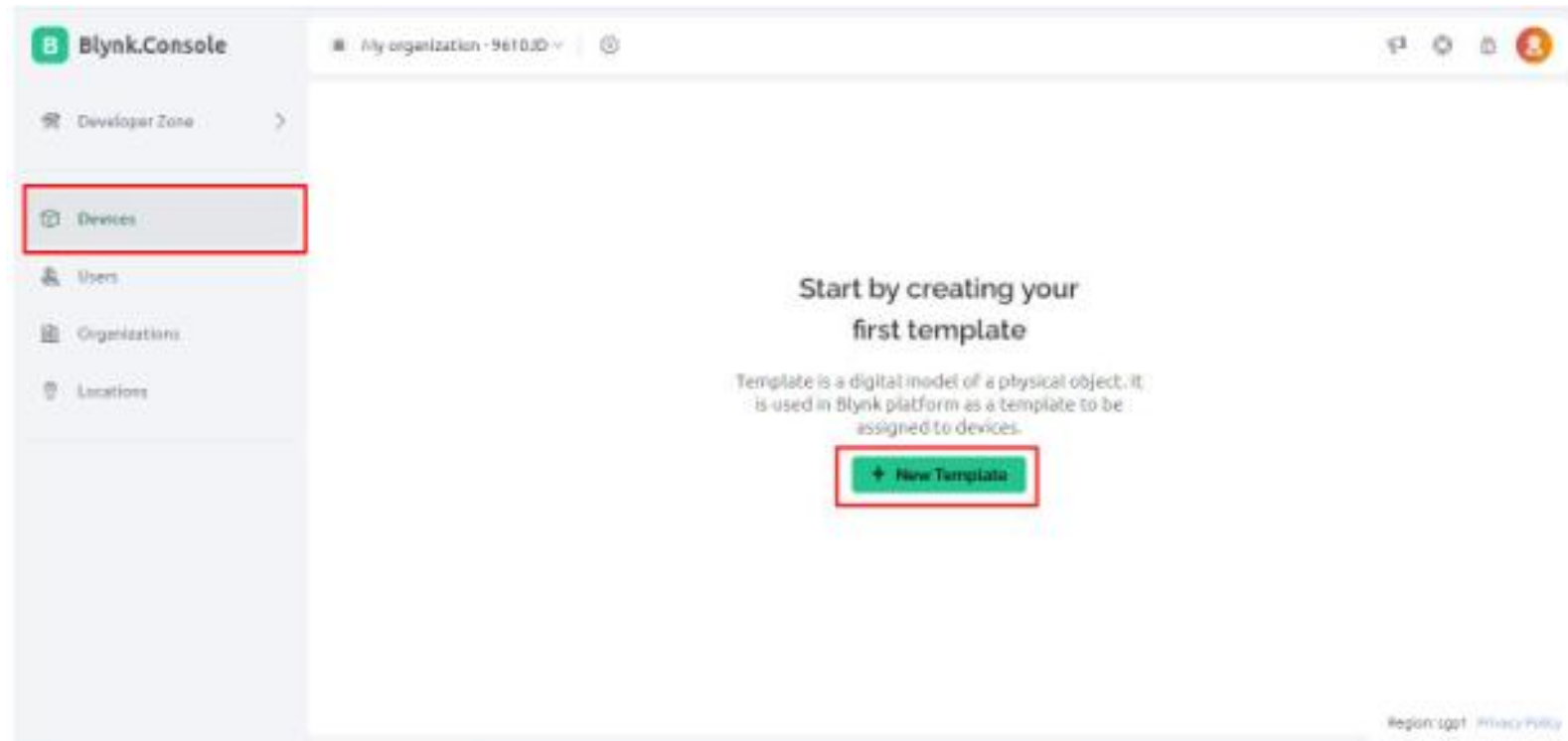
IoT Menggunakan Blynk

- Blynk adalah sebuah platform Internet of Things (IoT) yang memungkinkan pengembang untuk membuat aplikasi dan proyek IoT tanpa harus menulis banyak kode.
- Blynk dirancang untuk menjadi user-friendly dan memungkinkan integrasi perangkat keras dan perangkat lunak secara mudah.

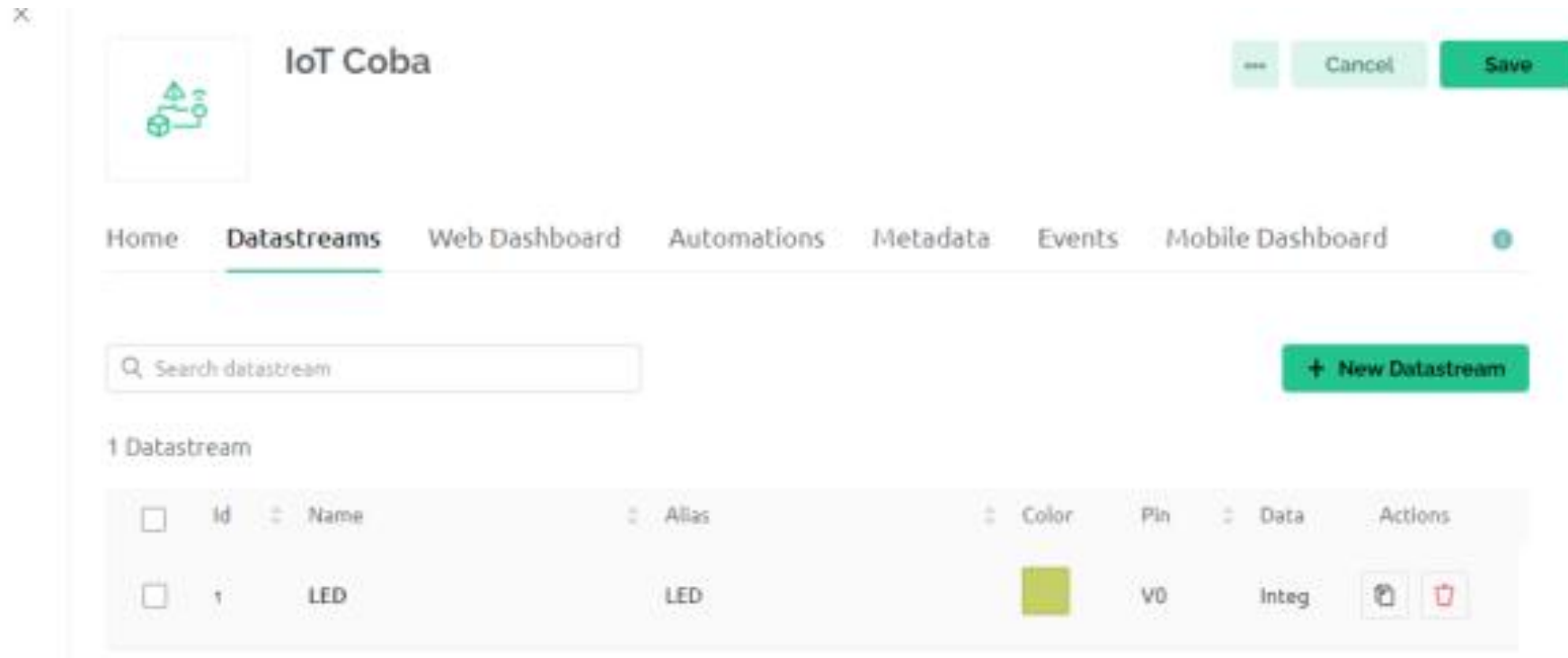


Langkah-Langkah Menggunakan Blynk

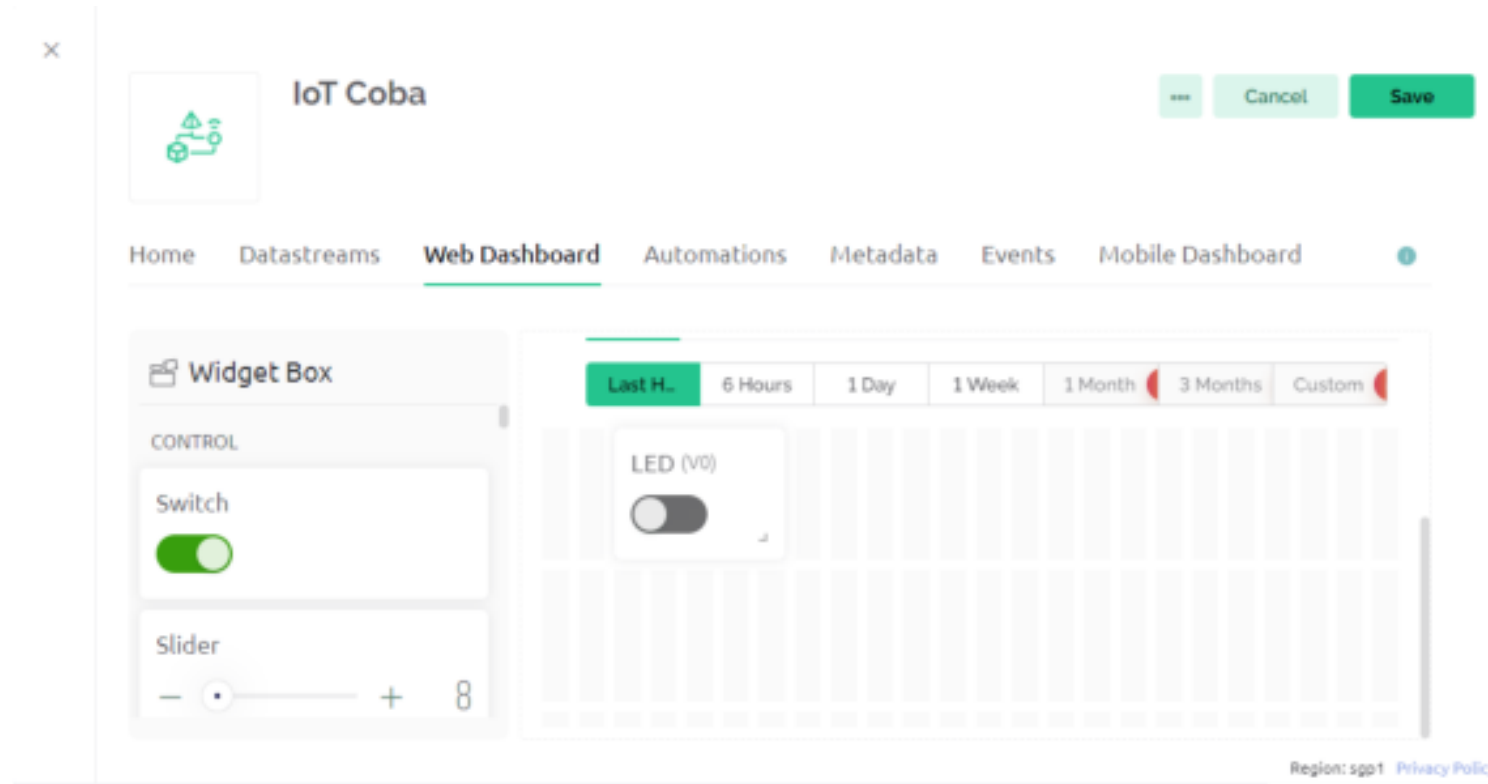
- Membuat Template Blynk



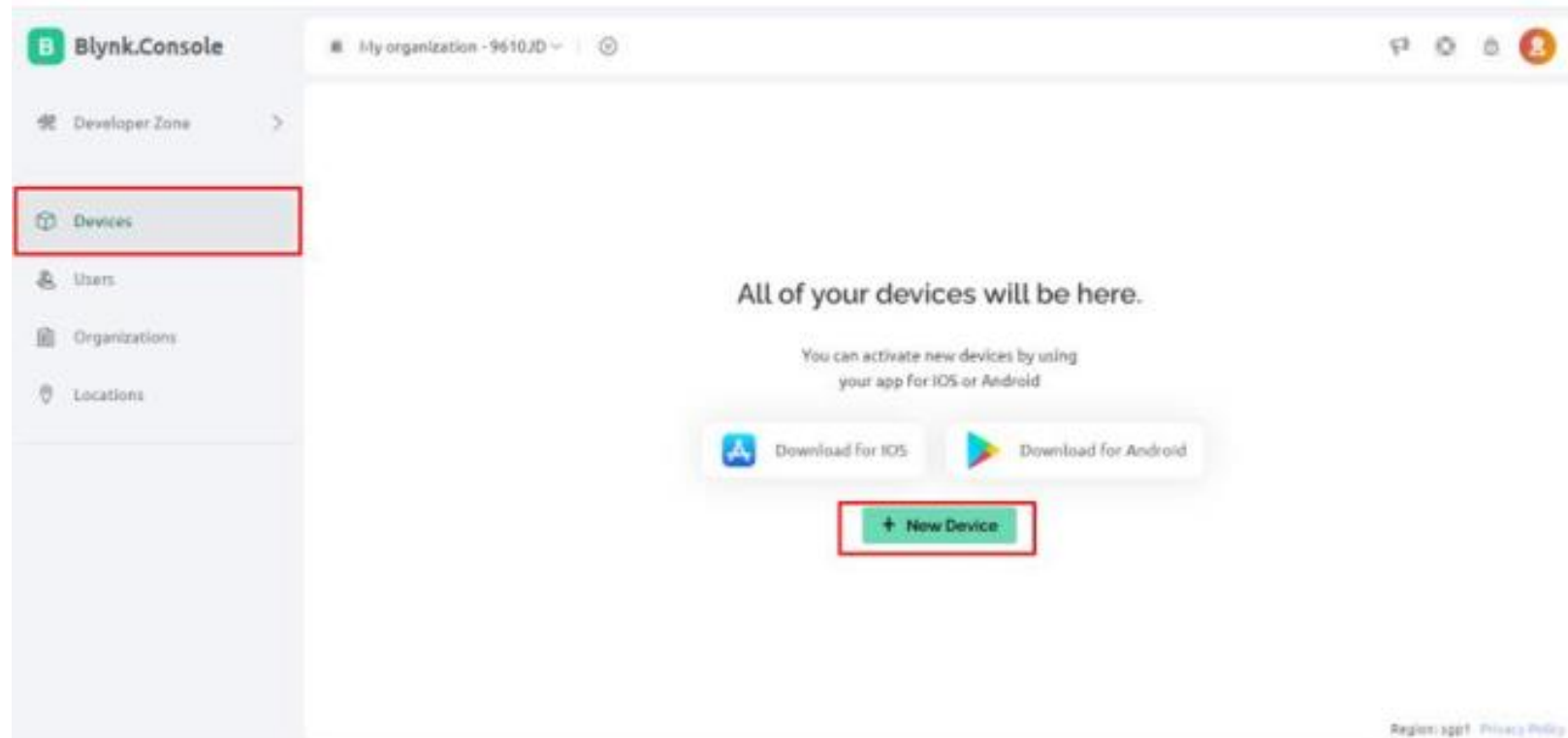
- Membuat Datastream



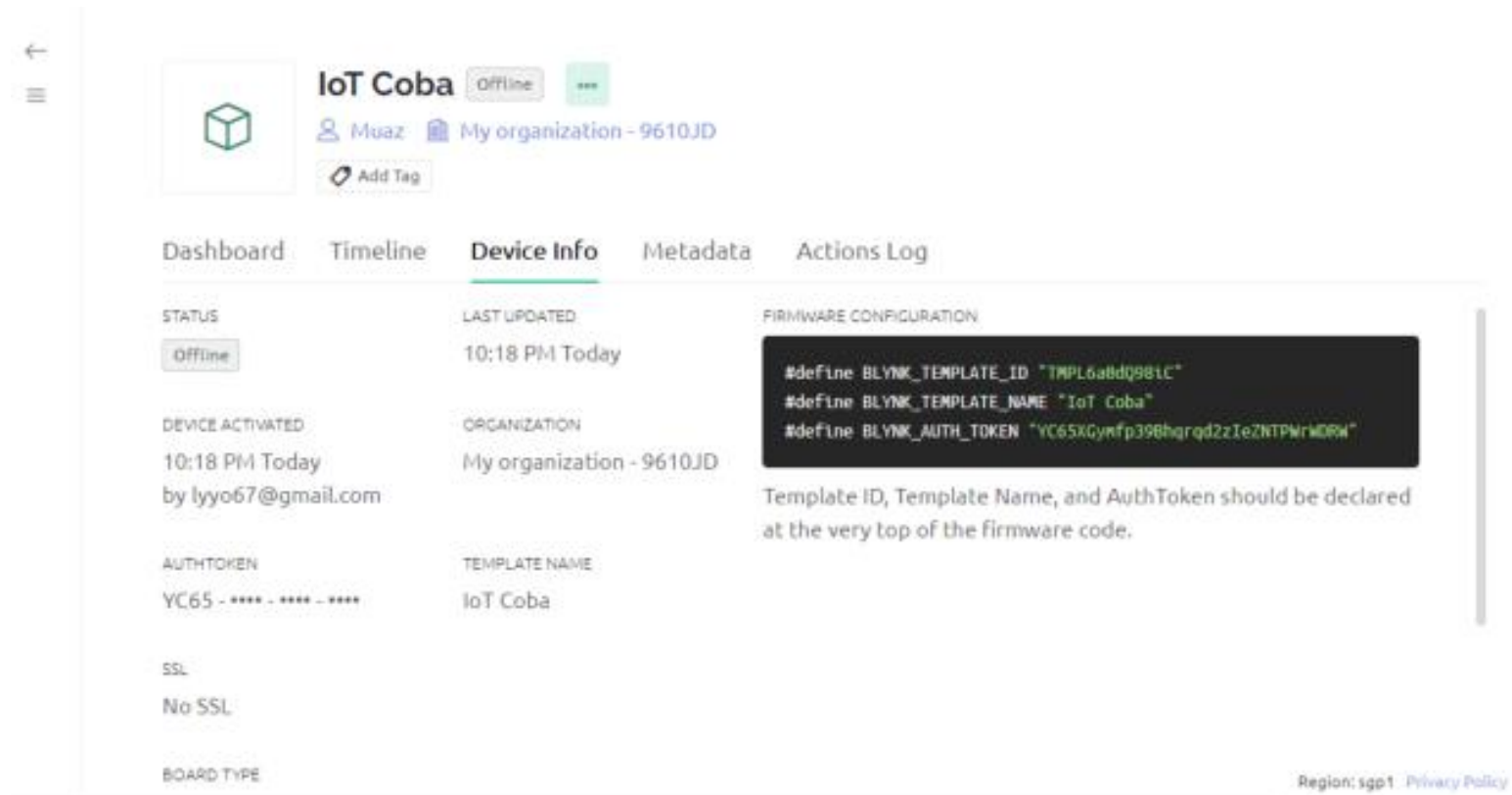
- Mendesain Web Dashboard



- Membuat Device baru



- Mengambil Firmware Configuration dari Device



The screenshot shows the 'IoT Coba' device page in a web interface. The device is currently 'Offline'. The page has tabs for 'Dashboard', 'Timeline', 'Device Info' (selected), 'Metadata', and 'Actions Log'. Under 'Device Info', there are sections for 'STATUS' (Offline), 'LAST UPDATED' (10:18 PM Today), 'DEVICE ACTIVATED' (10:18 PM Today by lyyo67@gmail.com), 'AUTH TOKEN' (YC65 - **** - **** - ****), 'SSL' (No SSL), and 'BOARD TYPE'. The 'FIRMWARE CONFIGURATION' section shows a code snippet with Blynk template ID, name, and auth token. A note below the code states: 'Template ID, Template Name, and AuthToken should be declared at the very top of the firmware code.' The footer shows 'Region: sgp1' and a 'Privacy Policy' link.

IoT Coba Offline

Muaz My organization - 9610JD

Add Tag

Dashboard Timeline **Device Info** Metadata Actions Log

STATUS
Offline

LAST UPDATED
10:18 PM Today

DEVICE ACTIVATED
10:18 PM Today
by lyyo67@gmail.com

ORGANIZATION
My organization - 9610JD

AUTH TOKEN
YC65 - **** - **** - ****

TEMPLATE NAME
IoT Coba

SSL
No SSL

BOARD TYPE

FIRMWARE CONFIGURATION

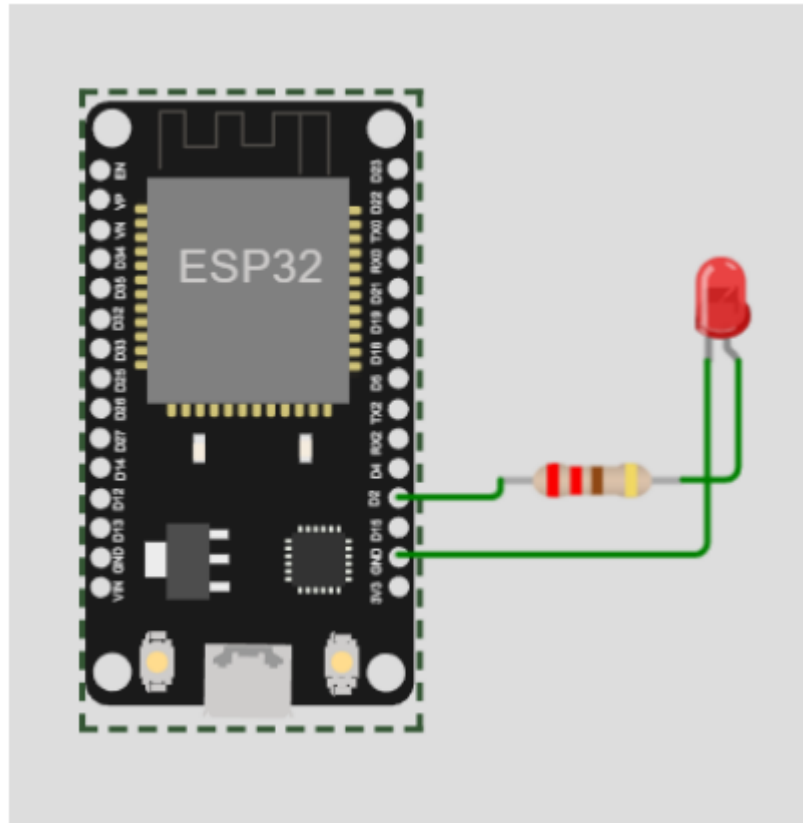
```
#define BLYNK_TEMPLATE_ID "TMPL6a8d0981C"  
#define BLYNK_TEMPLATE_NAME "IoT Coba"  
#define BLYNK_AUTH_TOKEN "YC65XGynfp396hgrqd2zIeZNTPWm0RW"
```

Template ID, Template Name, and AuthToken should be declared at the very top of the firmware code.

Region: sgp1 Privacy Policy

TUGAS

- Buatlah rangkaian berikut pada WOKWI, kemudian hubungkan ke Blynk
- Kumpulkan SS tampilan hasil running rangkaian pada WOKWI dan tampilan pada Blynk.
- Lengkapi dengan penjelasan cara kerja dan proses yang terjadi pada tiap langkah program, sertakan Nama dan NPM.



```
#define BLYNK_TEMPLATE_ID "TMPL6a0dQ98iC"
#define BLYNK_TEMPLATE_NAME "IoT Coba"
#define BLYNK_AUTH_TOKEN "YC65XGymfp39Bhqrqd2zIeZNTPrWDRW"
/* Comment this out to disable prints and save space */
#define BLYNK_PRINT Serial

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>

// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "wokwi-GUEST";
char pass[] = "";

// This function will be called every time slider
// widget in Blynk app writes values to the virtual pin 1
BLYNK_WRITE(V0)
{
  int pinvalue = param.asInt(); // assigning
  incoming value from pin V1 to a variable
  digitalWrite(2,pinvalue);
  // You can also use:
  // String i = param.asStr();
  // double d = param.asDouble();
  Serial.print("V0 slider value is: ");
  Serial.println(pinvalue);
}

void setup()
{
  // Debug console
  Serial.begin(9600);
  pinMode(2, OUTPUT);
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  // You can also specify server:
  //Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass,
  "blynk.cloud", 80);
  //Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass,
  IPAddress(192,168,1,100), 8080);
}

void loop()
{
  Blynk.run();
}
```


Contoh Hasil Running pada WOKWI dan Blynk:

